Efficiency in syllable structure: The trade-off between syllable complexity, phoneme inventory, and word length

The Menzerath-Altmann law states that the longer a linguistic unit becomes, the shorter its constituents (Altmann, 1980). This can be applied to words and syllables, i.e. for a given word, an increase in its overall length results in a decrease in the length of its component syllables. Existing research by Maddieson (2005, 2013) investigates the correlation between the syllable complexity of a language and the size of its consonant inventory; statistical evidence is presented suggesting these variables mutually reinforce each other. The present study relates these two sets of findings by investigating whether the direction and strength of Maddieson's positive correlation changes when word length is introduced as a covariate. A syllable here is defined as a compulsory nucleic vowel sound surrounded by optional consonant sounds in the onset and coda, allowing syllable complexity to be approximated using the mean number of intervocalic consonant clusters in a word. We trial a numerical measure of syllable complexity, as opposed to a categorical comparison based on the canonical pattern of a language used in previous work. Using multilingual word lists containing phonemic transcriptions of individual words and data regarding their phonological inventories, the set of ~2000 languages represented by both the Automated Similarity Judgement Programme (ASJP) database (Wichmann et al., 2025) and PHOIBLE (Moran & McCloy (eds), 2019) are tested for their correlation between approximated syllable complexity, mean word length, and phoneme inventory size. We predict that a language with a smaller phoneme inventory may rely on long words or complex syllables to achieve the same level of perceptual contrast as a language with a large phoneme inventory. Finally, we aim to determine a set of optimal tradeoff points between these three variables within a sample of natural languages using a three-dimensional Pareto frontier.

References:

Altmann, Gabriel. and Grotjahn, Rüdiger (1980). Prolegomena to Menzerath's law. *1980*, pp.1-10.

Maddieson, Ian (2005). Issues of phonological complexity: Statistical analysis of the relationship between syllable structures, segment inventories and tone contrasts. UC Berkeley PhonLab Annual Report, 1(1).

Maddieson, Ian (2013). Syllable Structure. In: Dryer, Matthew S. & Haspelmath, Martin (eds.) WALS Online (v2020.4) [Data set]. Zenodo. https://doi.org/10.5281/zenodo.13950591

Moran, Steven & McCloy, Daniel (eds.) (2019). PHOIBLE 2.0. Jena: Max Planck Institute for the Science of Human History. https://phoible.org/

Wichmann, Søren et al. (2025). The ASJP Database (version 21). https://asjp.clld.org/